## What is kube-proxy?

**Professional Answer**

kube-proxy manages networking rules on nodes to enable service-to-pod communication using iptables or IPVS.

**Unprofessional Answer**

kube-proxy handles networking.

## What is the role of kube-proxy in Kubernetes networking?

kube-proxy manages networking rules on worker nodes.
It enables communication between services and pods using iptables or IPVS.
kube-proxy provides service discovery and load balancing functionality.
It ensures traffic sent to a service IP is forwarded to the correct pod.
Without kube-proxy, Kubernetes services would not function correctly.

## kube-proxy (Networking Internals)

**What it does internally**

- Programs networking rules
- Uses:
    - iptables OR
    - IPVS (IP Virtual Server)

**Service Traffic Flow**

Client → Service IP → kube-proxy rules → Pod IP

**Modes**

- userspace (old)
- iptables (default)
- IPVS (high scale)

**Without kube-proxy**

Services won’t work

**What is kube-proxy?**

**Professional Answer**

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kube-proxy is a network component that runs on every worker node and manages Service networking.
It maintains network rules that allow Pods to communicate with Services.
It performs load balancing across backend Pods.
kube-proxy translates Service IPs into Pod IPs using iptables or IPVS.
It ensures stable networking abstraction in Kubernetes.

**Weak Answer**

kube-proxy is used for networking between pods.

**Why is kube-proxy needed?**

**Professional Answer**

Pods are ephemeral and their IPs change frequently.
kube-proxy provides a stable Service IP abstraction.
It dynamically updates routing rules when Pods are added or removed.
This allows applications to communicate reliably without knowing Pod IPs.
It enables service discovery and load balancing.

**Where does kube-proxy run?**

**Professional Answer**

kube-proxy runs as a DaemonSet on every worker node.
Each instance manages networking rules locally on that node.
It communicates with the API Server to watch Service and Endpoint changes.
It updates local routing rules accordingly.
This decentralized model improves scalability.